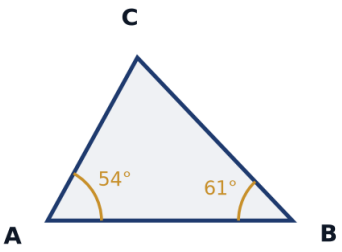
Transformations and congruence

Core Concepts: Transformations and congruence

1 In $\triangle ABC$ studying transformations and congruence, $m\angle A = 32^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?



a. 103°

b. 113°

c. 95°

d. 148°

2 A geometric figure scaled by factor $k = 2$ has an original side length of 6 cm. What is the corresponding image length?

a. 10 cm

b. 12 cm

c. 18 cm

d. 24 cm

3 On the coordinate plane, a line with slope $m = 3$ passes through $(0, 3)$. What is its y-coordinate when $x = 1$?

a. 3

b. 8

c. 6

d. 9

Procedural Fluency & Algebraic Manipulation

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4 In $\triangle ABC$ studying transformations and congruence, $m\angle A = 32^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 96°

b. 106°

c. 88°

d. 148°

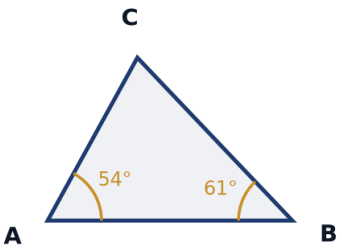
5 A geometric figure scaled by factor $k = 3$ has an original side length of 6 cm. What is the corresponding image length?

a. 15 cm

b. 18 cm

c. 24 cm

d. 54 cm



6 On the coordinate plane, a line with slope $m = 3$ passes through $(0, 3)$. What is its y-coordinate when $x = 2$?

a. 6

b. 11

c. 9

d. 12

Graphical & Geometric Interpretation

7 In $\triangle ABC$ studying transformations and congruence, $m\angle A = 32^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 89°

b. 99°

c. 81°

d. 148°

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8 A geometric figure scaled by factor $k = 4$ has an original side length of 6 cm. What is the corresponding image length?

a. 20 cm

b. 24 cm

c. 30 cm

d. 96 cm

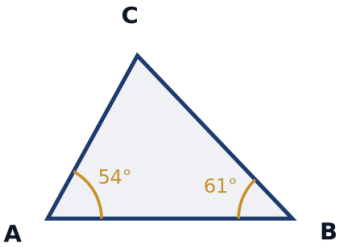
9 On the coordinate plane, a line with slope $m = 3$ passes through $(0, 3)$. What is its y-coordinate when $x = 3$?

a. 9

b. 14

c. 12

d. 15



Exam-Style Problem Solving & Modeling

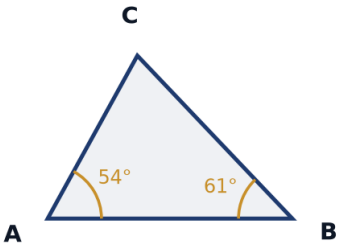
10 In $\triangle ABC$ studying transformations and congruence, $m\angle A = 32^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 82°

b. 92°

c. 74°

d. 148°



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11 A geometric figure scaled by factor $k = 5$ has an original side length of 6 cm. What is the corresponding image length?

a. 25 cm

b. 30 cm

c. 36 cm

d. 150 cm

12 On the coordinate plane, a line with slope $m = 3$ passes through $(0, 3)$. What is its y-coordinate when $x = 4$?

a. 12

b. 17

c. 15

d. 18

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